State of Health LPF18650

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Abstract

This project details the design, implementation and validation of an Edge AI-based State of Health (SoH) estimation system for lithium-ion batteries. The objective was to deploy optimized deep learning architectures directly onto a resource-constrained microcontroller, enabling real-time degradation monitoring without relying on cloud infrastructure.

The system's core features a validation setup that couples an STM32F401RE microcontroller with a host PC streaming real battery cycling data. The STM32 serves as the embedded processing unit, dedicated to handling sample-by-sample data acquisition, buffering a 128-step sliding window, performing onboard Z-score normalization, and executing quantized neural network inference via the ST X-CUBE-AI runtime framework. Both a 1D Convolutional Neural Network (1D-CNN) and a Temporal Convolutional Network (TCN) were trained, converted to ONNX, and quantized through Post-Training Quantization (INT8 PTQ) to evaluate their trade-offs in accuracy, memory footprint, and latency.

The quantized models execute well within the microcontroller's Flash and RAM limits while maintaining good accuracy, validating the possibility of deploying deep temporal networks for onboard battery diagnostics.

1 Introduction

Problem Statement

Lithium-ion batteries based on Graphite/Lithium Iron Phosphate (LFP) chemistry are widely adopted in electric vehicles (EV) and stationary storage due to their thermal stability, long cycle life, and cost-effectiveness. However, accurate State of Health (SoH) estimation remains a major operational challenge, particularly when assessing cells across both first-life automotive cycling and second-life repurposing. As highlighted in recent long-term ageing studies battery degradation is highly non-linear. While cloud-based deep learning methods offer superior estimation accuracy, they incur substantial transmission latency, bandwidth costs, and communication dependencies. The challenge lies in developing an autonomous, edge-deployed inference system capable of executing lightweight temporal deep learning models directly on low-power microcontrollers to track continuous degradation patterns in real time.

Objectives

The main objective of this work was to design, deploy, and experimentally validate an Edge AI-based State of Health (SoH) monitoring system capable of:

- Performing real-time degradation tracking of commercial lithium-ion cells by directly processing dynamic voltage and current time-series signals.
- Executing embedded deep learning inference on a highly resource-constrained ARM Cortex-M4 microcontroller (STM32F401RE), evaluating and comparing both 1D Convolutional Neural Network (1D-CNN) and Temporal Convolutional Network (TCN) topologies quantized to 8-bit integer precision (INT8 PTQ).
- Implementing autonomous onboard data pre-processing, including local sliding-window accumulation (128 samples) and hardware-efficient Z-score standardization directly inside the microcontroller firmware.
- Establishing a reliable communication pipeline between a host PC and the target microcontroller using an ASCII-based streaming protocol over UART.

By focusing on memory efficiency, computational determinism, and cost-effectiveness, the project aims to demonstrate that accurate and autonomous battery health diagnostics can be deployed on standard, low-cost microcontrollers.

2 Background on State of Health (SoH) Estimation

The State of Health (SoH) is an essential metric in Battery Management Systems (BMS) that quantifies the remaining capacity of a battery cell relative to its initial, nominal capacity.

$$\text{SoH} = \frac{C_{\text{actual}}}{C_{\text{nominal}}} \times 100\% \tag{1}$$

Where:

- C_{actual} is the remaining discharge capacity of the cell measured under reference conditions (Ah).
- C_{nominal} is the initial rated capacity specified by the manufacturer (1.2 Ah for the 18650 cell).

In electric vehicle (EV) applications, cells typically operate during their "first life" until their SoH drops to approximately 80% to 70%. Below this threshold, batteries are commonly retired from vehicles and repurposed for "second-life" stationary energy storage rather than being immediately recycled.

3 Dataset

The empirical foundation of this study relies on a comprehensive Lithium-Iron-Phosphate/Graphite (LFP/Gr) battery degradation dataset, established through a collaborative research initiative between Iowa State University and Iowa Lakes Community College.

The experimental campaign was carried out on 64 commercial cylindrical 18650 high-power cells with a rated nominal capacity of 1.2 Ah. To systematically capture a broad spectrum of degradation mechanisms and operational behaviors, the cells were partitioned into 11 distinct cycling cohorts, comprising six cells per group for cohorts 1 through 10, and four cells for cohort 11.

Degradation tracking was performed via periodic Reference Performance Tests (RPTs) conducted every 3.5 days, corresponding to roughly 100 equivalent full cycles. These characterisation routines served to record the remaining full discharge capacity and track internal resistance variations.

The dataset tracks continuous cell health degradation from a pristine baseline (SoH = 1.0) down to the standard End-of-Life (EoL) threshold, defined as an SoH of 0.8 (representing 80% retained nominal capacity). Due to the heterogeneous thermal and mechanical stress profiles, cell degradation trajectories exhibit significant diversity, with operational lifetimes to EoL ranging from 2,400 cycles for the most demanding cycling conditions to 9,000 cycles for the least aggressive regimes.

4 Methodology

The development of the Edge AI-based State of Health (SoH) estimation system followed a structured methodology combining offline deep learning design, embedded firmware implementation and real-time validation.

The engineering process was partitioned into sequential phases: data pre-processing and dynamic window extraction, baseline model definition and training in floating-point precision, post-training integer quantization (INT8 PTQ), target firmware integration on the STM32 board, and systematic end-to-end latency and accuracy verification through a streaming interface.

Tools and Development Environment

A variety of software tools and environments were employed to support each stage of the project:

- **Python:** used for managing data, feature normalization, sliding-window dataset generation, and defining the neural network architectures (1D-CNN and TCN), as well as executing the training loops and performance benchmarking.
- **ONNX (Open Neural Network Exchange):** used as the open intermediate representation format to serialize the trained models, enabling graph optimization prior to embedded deployment.
- **ST X-CUBE-AI:** an expansion package within the STM32Cube ecosystem used for model compression, Post-Training Quantization (INT8 PTQ) and static memory allocation analysis.
- **STM32CubeMX:** used for hardware peripheral initialization and configuring UART interrupt routines and Direct Memory Access (DMA) channels.
- **STM32CubeIDE:** the development environment used for C/C++ firmware development.

Dataset Construction and Preprocessing Pipeline

To train, evaluate, and deploy the machine learning models (1D-CNN and TinyTCN) for Battery State of Health (SoH) estimation, a dedicated data processing pipeline was developed. The objective of this pipeline is to transform raw reference performance test (RPT) data and continuous cycling time series into normalized, windowed tensor representations suitable for deep neural networks.

The preprocessing pipeline is divided into three fundamental stages: Ground Truth extraction, feature filtering and normalization, and temporal sliding window generation.

1. Ground Truth SoH Extraction and Interpolation

The reference ground truth for capacity degradation is derived from the periodic Reference Performance Test (RPT) files. For each battery cell, the maximum discharge capacity ($C_{\max}$) is extracted during the reference discharge segments, allowing the instantaneous State of Health to be evaluated against the battery's nominal capacity as

$$\text{SoH} = \frac{C_{\max}}{C_{\text{nominal}}}. \tag{2}$$

Furthermore, since RPT tests are conducted at discrete weekly intervals, the cycling time series originally lack direct SoH labels for every second. To overcome this limitation, the discrete RPT records are merged with the cycling data and continuously propagated across all raw time-steps using linear interpolation.

2. Feature Filtering and Zero-Leakage Normalization

The input features consist of raw time-series measurements, specifically Voltage (V) and Current (I). High-frequency measurement noise is first attenuated by applying a 5-point causal moving average filter directly to the voltage and current series. To strictly prevent data leakage between training and evaluation stages, ten specific cells (05, 12, 18, 25, 33, 41, 48, 52, 59, 63) are completely isolated to form the independent test set. Standard Z-score normalization parameters (μ_V, σ_V, μ_I, σ_I) are then computed strictly from the training cells using the standard scaling transformation:

$$\hat{x} = \frac{x - \mu_x}{\sigma_x}. \tag{3}$$

These computed global mean and standard deviation metrics are permanently stored within the final output dataset archives to enforce identical feature scaling during ONNX static quantization and embedded deployment on the STM32 microcontroller.

3. Time Series Windowing and Output Packaging

To properly format the continuous signals for 1D convolutional and temporal architectures, a sliding window mechanism is applied to the normalized time series. Using a fixed window size of 128 samples and a non-overlapping stride of 128, each continuous signal segment is restructured into an input tensor matrix of shape $(128, 2)$, representing 128 consecutive timesteps of normalized $[V, I]$ pairs. Each generated window is then paired with a single ground truth target, defined as the continuous, interpolated SoH value corresponding precisely to the final timestep of that specific temporal window.

Overall, across the total population of 64 cells, isolating 10 cells for testing allocates approximately 15% of the dataset to the test set, while the remaining 54 cells are split into 70% for training and 15% for validation, guaranteeing that all evaluation metrics reflect genuine generalization on unseen battery specimens.

Architecture Definition

The system is built around an STM32F401RE NUCLEO development board, equipped with 512 KB of Flash memory and 96 KB of SRAM. The STM32 serves as the central embedded inference engine, responsible for maintaining a local sliding window, performing onboard feature standardization, and executing optimized deep learning models.

To evaluate real-time performance a host PC acts as an emulated Battery Management System (BMS) sensing unit. The host platform replays real-world cycling data and streams individual discrete measurements to the microcontroller over a serial interface.

Upon receiving each measurement pair, the STM32 validates frame integrity and buffers the data into a 128-step temporal sliding window. When the window is fully populated, the microcontroller triggers an onboard inference pass using the deployed 8-bit quantized neural network. The resulting SoH estimation is formatted and transmitted back to the host computer for visual monitoring, and latency evaluation.

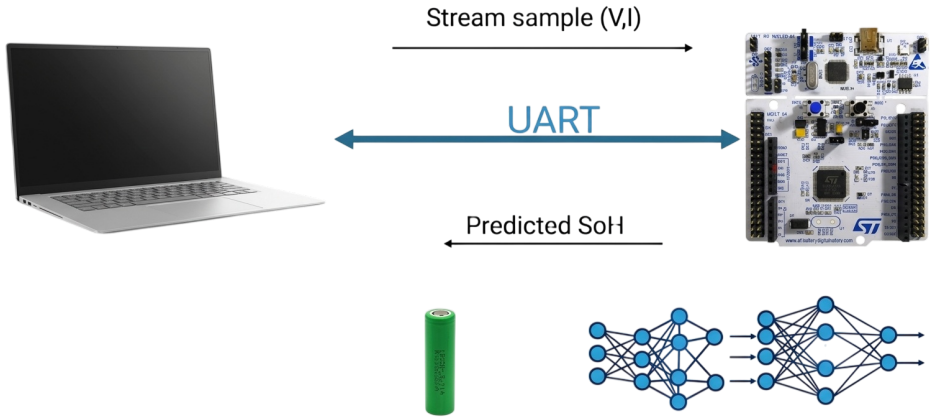


Figure 1: system architecture scheme

UART communication

To establish a reliable data interchange between the host computer and the microcontroller a UART communication was implemented, operating at a standard baud rate of 115,200 bps with an 8-bit data word.

The interaction begins with a synchronous parameter initialization sequence designed to align feature scaling constants. Before transmitting cycling measurements, the host script broadcasts a configuration string containing the statistical values V_{mean}, V_{std}, I_{mean} and I_{std} computed during training.

Following a successful handshake confirmation of a correct communication, the host starts an emulated BMS sensor streaming loop, sequentially dispatching discrete voltage and current measurements. On the microcontroller, the received ASCII is converted into float pairs and appended directly into dedicated voltage and current sliding-window arrays.

Once the microcontroller accumulates 128 consecutive sensor pairs, the sliding window is locked, and the onboard execution sequence is triggered. The stored time-series is standardized using the pre-loaded Z-score parameters and mapped onto the INT8 input

quantization grid of the neural network. Then the deep learning model is executed and the resulting 8-bit integer prediction is de-quantized into a floating-point State of Health value bounded between 0.0 and 1.0. The microcontroller formats this output as a response message and transmits it back across the UART channel, and the sliding-window buffer counter is reset to zero to ingest subsequent cycling frames without interruption.

Firmware development

The firmware was developed following a step-by-step methodology to guarantee a reliable real-time execution.

The runtime logic operates across three functional phases:

1. **System Initialization and Dynamic Handshaking:** The microcontroller initializes the model parameters and clears the internal state of a 31-point circular Moving Average Filter (MAF). It then halts continuous acquisition until receiving a configuration string containing the Z-score statistics (V_{mean}, V_{std}, I_{mean}, I_{std}). These values are stored directly in volatile registers, prompting the microcontroller to return an ACK token to authorize streaming.

2. **Window Buffering and Local Feature Scaling:** As streaming begins, discrete voltage and current measurements are extracted and accumulated into independent 128-element time-series buffers. When the window counter reaches capacity, input scaling is performed on-chip: each raw sample is normalized according to the active Z-score constants and quantized into signed 8-bit integers using the network's specific quantization scale and zero-point parameters.

3. **Inference Execution and Moving Average Filtering:** The ST X-CUBE-AI inference runtime processes the populated input tensor, yielding a single INT8 output activation that is dequantized into a floating-point State of Health value. To filter high-frequency noise, this prediction is pushed into the circular moving average buffer, computing a moving mean. Both the raw and filtered SoH values are decomposed into fixed-point integer representations and returned via UART.

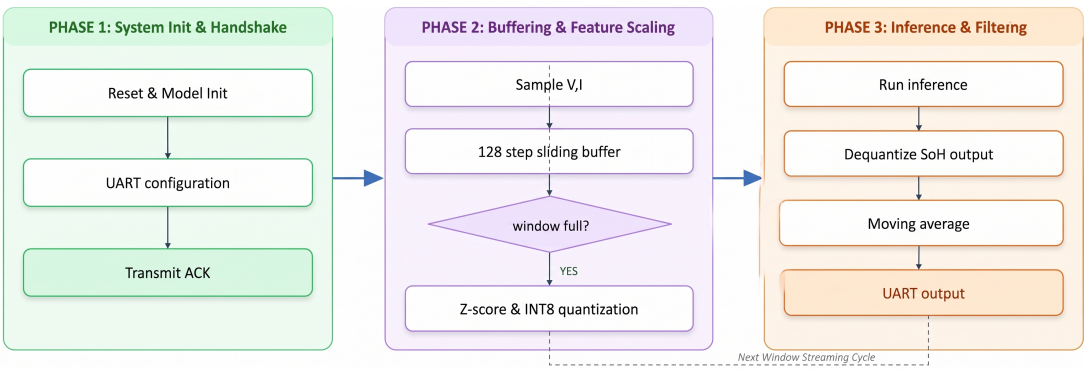


Figure 2: firmware functionality scheme

5 Experimental results

The experimental validation of the proposed State of Health (SoH) tracking framework followed a comparative strategy. An initial baseline architecture, based on a one-dimensional Convolutional Neural Network (1D-CNN), was designed, trained, and evaluated. Subsequently, to better capture the extended temporal dependencies present in battery degradation sequences, a Temporal Convolutional Network (TCN) was implemented. Because the TCN achieved superior predictive accuracy while remaining strictly within the hardware resource constraints of the target microcontroller, it was ultimately selected for the final embedded deployment.

5.1 Preliminary Baseline: 1D-CNN Architecture and Evaluation

The primary baseline model was formulated as a compact 1D-CNN structured to extract localized temporal features directly from the two-channel input tensor consisting of voltage and current series spanning a 128-step temporal horizon.

Architectural Specification

The baseline 1D-CNN was designed with a simple structure to satisfy the memory and computing constraints of the microcontroller. The network takes as input a sequence of 128 samples with two distinct channels corresponding to the measured voltage and current signals.

The feature extraction stage consists of four consecutive one-dimensional convolutional layers with a progressively increasing number of filters, specifically set to 16, 32, 32, and 64. Each convolutional operation applies a kernel size of 3 with symmetric padding to maintain sequence boundaries, followed directly by a ReLU activation function to introduce non-linearity. To gradually reduce the temporal dimension without significantly increasing computational cost, average pooling layers are placed between the convolutional stages. After the fourth convolutional layer, an adaptive average pooling layer condenses the entire remaining time axis into a compact 64-element feature vector.

Finally, this 64-element descriptor is fed into a small regression head composed of a fully connected layer with 64 hidden units, a ReLU activation, and a dropout layer. The network ends with a single linear output neuron that directly predicts the State of Health value.

1D-CNN Performances

The baseline 1D-CNN model was trained and subsequently tested on unseen cycling cells to evaluate its generalization ability across unseen battery degradation dynamics. Model performance was quantified using three standard statistical metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Evaluation Metric	1D-CNN Performance
Mean Absolute Error (MAE)	0.0242
Root Mean Square Error (RMSE)	0.0325
Mean Absolute Percentage Error (MAPE)	2.61%

As evidenced by the results, the 1D-CNN demonstrated satisfactory predictive capability. However, because the pooling layers compress the data by averaging nearby values, they lose part of the precise chronological sequence of the signals. As a result, the 1D-CNN tended to produce small fluctuations in its predictions.

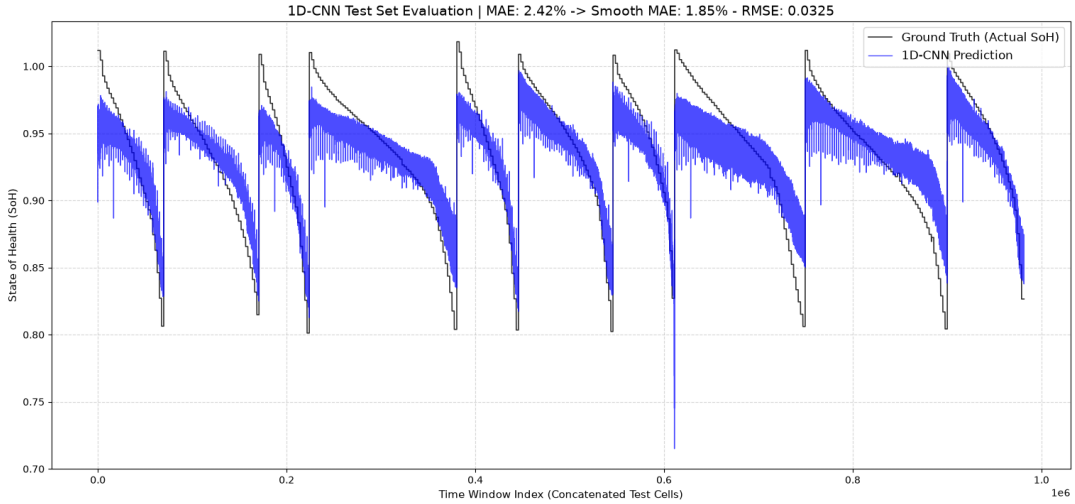


Figure 3: Evaluation of the baseline 1D-CNN on the test dataset across concatenated unseen battery cells.

Figure 3 illustrates the inference results of the baseline 1D-CNN compared against the reference ground truth across several unseen test cells concatenated along the temporal window index.

The raw output predictions (blue) exhibit pronounced high-frequency noise and local oscillations around the true State of Health trajectory (black). These transient fluctuations are primarily caused by the dynamic current and voltage variations within each cycle, where the pooling layers of the 1D-CNN struggle to resolve fine-grained temporal ordering. By applying a moving average filter to post-process the predictions, the smoothed response significantly mitigates these rapid deviations, reducing the MAE to 1.85%. This notable improvement through smoothing confirms that while the 1D-CNN effectively captures global aging trends, it lacks the temporal fidelity required to reject short-term dynamic disturbances. To overcome this limitation a Temporal Convolutional Network (TCN) was investigated.

5.2 Proposed Model: Temporal Convolutional Network (TCN)

To overcome the architectural limitations of conventional pooling operations and capture multi-scale temporal dynamics across extended sequences, a Temporal Convolutional Network (TCN) was designed and implemented as the primary candidate for embedded execution.

Architectural Specification

The network architecture is organized into a modular pipeline combining dilated residual convolutional blocks with a compact regression head. The input layer accepts the standardized two-channel temporal sequence of length 128 corresponding to the measured voltage and current dynamics.

The feature extraction backbone consists of four stacked TemporalBlock stages with channel dimensions configured as 16, 32, 32, and 64. Within each temporal block, two consecutive one-dimensional dilated convolutional layers (Conv1d) are applied using a kernel size of $k = 5$. An exponentially increasing dilation factor is assigned to successive blocks, allowing the effective receptive field to expand across the full sequence. Each convolutional operation is followed by Batch Normalization, a ReLU activation function, and a dropout layer.

Following the final residual block (64 feature channels), an Adaptive Average Pooling layer aggregates the temporal features into a fixed-length 64-dimensional descriptor. This vector is flattened and fed into a dense regression head consisting of a linear projection to 32 hidden units, a ReLU non-linearity, a regularizing dropout layer and a final linear layer that outputs the scalar State of Health estimate.

TCN Performances

The trained TCN was evaluated on the same benchmark of unseen cycling cells compared to the baseline 1D-CNN. Model accuracy was assessed via Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Evaluation Metric	Baseline 1D-CNN	Proposed TCN
Mean Absolute Error (MAE)	0.0242 (2.42%)	0.0230 (2.30%)
Root Mean Square Error (RMSE)	0.0325	0.0309
Mean Absolute Percentage Error (MAPE)	2.61%	2.49%

As for the 1D-CNN, we operate a test against the reference ground truth across several unseen test cells concatenated along the temporal window index.

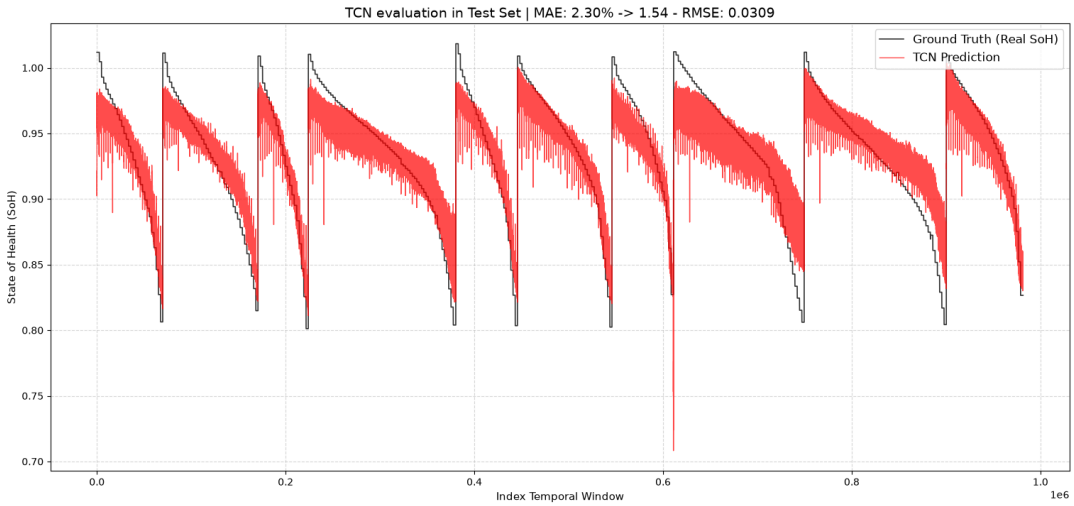


Figure 4: Evaluation of the TCN on the test dataset across concatenated unseen battery cells.

Figure 4 shows the evaluation results of the proposed TCN architecture, which demonstrates superior trajectory tracking and higher estimation accuracy compared to the baseline 1D-CNN model.

The TCN architecture leverages dilated causal convolutions, which expand its receptive field across the 128-sample time window without losing temporal order. As a result, the TCN predictions follow the non-linear battery capacity fade curve much more closely, particularly in the central degradation phase of each cell, whereas the 1D-CNN exhibits larger variance and periodic underestimation during early cycle life. TCN demonstrates higher stability across all test cells, maintaining tighter prediction boundaries around the ground truth profile.

In conclusion, the TCN model proves to be a more robust candidate for edge deployment, providing superior baseline FP32 accuracy and a smoother estimation curve.

Results after quantization

The static quantization pipeline was implemented using the ONNX Runtime quantization toolchain. The model was converted from 32-bit Floating-Point (FP32) to an 8-bit Signed Integer (INT8) representation for both weights and activations.

The performance evaluation was conducted on the full test dataset, comparing the original PyTorch FP32 model against the quantized ONNX INT8 model.

The primary metrics evaluated are the same as before measured on raw predictions.

Metric	TCN FP32	TCN INT8 (PTQ)	Absolute Degradation (Δ)
MAE	0.0230	0.0516	+0.0286
RMSE	0.0309	0.0691	+0.0382
MAPE (%)	2.49%	5.51%	+3.02%

Deployability Verdict

While the FP32 model provides a promising starting point for State of Health estimation, its accuracy can certainly be further refined in future iterations. On the other hand, the Post-Training Quantization (PTQ) results currently fall short of target performance expectations, as the error degradation (ΔMAE $\simeq 2.86\%$, leading to an absolute MAPE of 5.51%) represents a non-negligible loss in precision. Consequently, future work should focus on investigating more sophisticated quantization strategies, like Quantization-Aware Training (QAT) or mixed-precision approaches, to minimize this gap.

Nevertheless, the hardware deployment itself remains a strong proof of concept: the transition to the STM32 microcontroller yields substantial gains in execution speed, significantly reduced power consumption, and guaranteed deterministic latency, proving the practical feasibility of real-time edge processing for Battery Management Systems (BMS).

5.3 Evaluation of Frequency-Domain Features

An alternative configuration was evaluated by transforming the input time-series signals (V and I) into the frequency domain using the Fast Fourier Transform (FFT) prior to model training. The rationale was to investigate whether spectral components and harmonic distribution could provide a more compact representation for State of Health (SoH) estimation.

However, the experimental results obtained with the FFT-based approach were noticeably inferior compared to those achieved by the temporal architectures (1D-CNN and TinyTCN). Some possible causes include:

1. **Loss of Temporal Phase and Sequence Dynamics:** Applying the FFT converts the 128-sample time window into frequency spectra, completely discarding the explicit temporal sequencing of transient events.
2. **Sub-optimal Feature Representation for Slow Degradation:** State of Health is a slow-varying parameter governed by low-frequency energy distributions. The resulting FFT representations were dominated by DC components and near-zero frequency bins, rendering high-frequency spectral coefficients mostly noisy and uninformative for regression.

Consequently, the frequency-domain pipeline was discarded in favor of direct time-domain processing, which preserves the full dynamic profile of the battery signals and yields superior estimation accuracy.

6 Conclusion

This work has demonstrated the successful design, embedded deployment, and experimental validation of an Edge AI-based State of Health (SoH) estimation system for lithium-ion

batteries. By combining an STM32F401RE microcontroller with deep learning models, the prototype verified the feasibility of executing continuous temporal inference directly on edge hardware without reliance on cloud infrastructure. The implemented pipeline reliably handled discrete voltage and current streaming, executed deterministic on-chip normalization, and delivered real-time health predictions while adhering to the strict timing constraints of a modern Battery Management System (BMS).

Despite these achievements, the current implementation exhibits some limitations. While the floating-point TCN baseline achieved high predictive fidelity (MAE: 2.30%, RMSE: 0.0309), the Post-Training Quantization (PTQ) to INT8 precision led to a noticeable performance degradation, increasing the MAE to 5.16% and the MAPE to 5.51%. This precision penalty stems from cumulative rounding noise across consecutive convolutional blocks, which introduce a slight offset in the output layer, penalizing the predicted State of Health (SoH) trajectory.

Despite this precision penalty, the experimental deployment on the STM32F401RE provides a proof of concept for edge battery analytics. Executing inference locally delivers rapid cycle times, minimal power draw, and strictly deterministic latency, validating the practical viability of embedding deep temporal models directly within Battery Management Systems (BMS).

References

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